Aayushi Malhotra

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EXPERIENCE

IDX Exchange LLC — Oct 2025 – Present
Software Engineering Intern — Remote

- Built Python + SQL ingestion checks for listing feeds (schema validation, dedupe, null thresholds, idempotent loads), reducing bad records reaching search and reporting by about 35%.
- Implemented feed health KPIs and dashboards (freshness, reject reasons, coverage, deltas) with documented definitions and scheduled refreshes, cutting ad hoc investigations by roughly 2 to 3 hours per week.
- Tuned MySQL schemas and query plans for high-traffic filters (index design, EXPLAIN profiling, query rewrites), lowering median latency about 30% and improving tail behavior under peak load.

Ericsson — May 2024 – Jul 2024
Cloud Engineering Intern — Remote

- Refactored Apache Beam pipelines on Google Cloud Dataflow to transform telemetry into analytics tables (windowing, combiner efficiency, IO tuning), improving end-to-end processing time about 30% for release reporting.
- Added validation probes, structured logs, and rollout checks; reduced time to isolate data regressions by about 25% by making failures reproducible from logs and queries.

Michigan State University, College of Social Science — Mar 2024 – Feb 2025
Undergraduate Research Assistant — East Lansing, MI

- Built analysis-ready datasets from public APIs and structured records using Python (pandas) with schema guards and deduping; trained baseline models with error analysis and reproducible reporting artifacts.

Innefu Labs — May 2023 – Jul 2023
Data Analytics Intern — Gurgaon, India

- Standardized PySpark ingestion and preprocessing for high-volume security telemetry (partitioning, caching, vectorized transforms), reducing batch processing time about 25% and improving dataset readiness.
- Delivered Streamlit dashboards and scheduled reports with consistent metric definitions, cutting manual reporting effort about 30% through self-serve filters and repeatable refresh runs.

ByteBlanket — May 2022 – Jul 2022
Machine Learning Intern — Dubai, UAE

- Built an NLP sentiment classifier in Python (spaCy, scikit-learn) with evaluation and data checks; automated 50%+ of support triage and improved throughput about 35% in pilot workflows.

PROJECTS

FlakeWatch, CI Flakiness and Failure Analytics

- Built a FastAPI service that ingests CI runs into PostgreSQL and computes per-test reliability metrics (flake rate, trend deltas, failure clustering) across hundreds of tests with auditable SQL.
- Shipped a React + TypeScript triage dashboard (top flaky, regressions, per-test history) and CI automation via GitHub Actions (migrations, pytest, build) to tighten debug loops.

MeterStack, Event-Based KPI Reporting

- Implemented idempotent ingestion endpoints and schema validation for product events, producing clean fact tables and automated exports at thousands of events per day for recurring reporting.
- Built cohort and plan-level KPI views with standardized logic, reducing time spent reconciling conflicting dashboards and improving consistency across weekly reviews.

HackDuke Code for Good (Prize Winner) — Feb 2025
Built a data reliability and analytics prototype with validation checks and a stakeholder dashboard to turn noisy partner inputs into a small set of actionable signals.

SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Bash
Backend & APIs	FastAPI, REST APIs, Data Validation, Background Jobs, Query Optimization
Data & Storage	PostgreSQL, MySQL, Pandas, NumPy, PySpark, ETL, Data Modeling
Infra & Tooling	Git, GitHub Actions, Docker, Kubernetes, Linux, GCP Dataflow (Beam), Monitoring/Logging

EDUCATION

Michigan State University, East Lansing, MI — Expected May 2026
Bachelor of Science, Computer Science; Minor in Cognitive Science